



Changing Trends in Long-Term Sentiments and Neighborhood Determinants in a Shrinking City

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Yunmi Park¹ , Minju Kim¹, Jiyeon Shin¹ ,
and Megan E. Heim LaFrombois² 

Abstract

This research examined social media's role in understanding perceptions about the spaces in which individuals interact, what planners can learn from social media data, and how to use social media to inform urban regeneration efforts. Using Twitter data from 2010 to 2018 recorded in one U.S. shrinking city, Detroit, Michigan, this paper longitudinally investigated topics that people discuss, their emotions, and neighborhood conditions associated with these topics and sentiments. Findings demonstrate that neighborhood demographics, socioeconomic, and built environment conditions impact people's sentiments.

Keywords

shrinking cities, urban regeneration, topic modeling, sentiments, big data, Twitter, Detroit

Abstract

Esta investigación examinó el papel de las redes sociales en la comprensión de las percepciones sobre los espacios en los que los individuos interactúan, lo que los planificadores pueden aprender de los datos de las redes sociales y cómo usar las redes sociales para informar los esfuerzos de regeneración urbana. Utilizando datos de Twitter de 2010 a 2018 registrados en una ciudad en proceso de contracción de Estados Unidos, Detroit, Michigan, este documento investigó longitudinalmente temas que las personas discuten, sus emociones y las condiciones del vecindario asociadas con estos temas y sentimientos. Los hallazgos demuestran que la demografía del vecindario, las condiciones socioeconómicas y del entorno construido afectan los sentimientos de las personas.

Keywords

Ciudades en contracción, regeneración urbana, modelado de temas, sentimientos, “Big Data”, Twitter, Detroit

摘要

摘要: 这项研究考察了社交媒体在理解个人互动空间中的角色, 规划者可以从社交媒体数据中学到如何使用社交媒体为城市新建工作提供信息方面的作用。本文使用 2010 年至 2018 年记录在美国密歇根州底特律市的推特数据, 纵向调查了人们讨论的话题、他们的情绪以及与此话题和情绪相关的社区状况。 调查结果表明, 社区人口统计、社会经济和建筑环境条件会影响人们的情绪。

关键词

关键词, 收缩城市, 城市更新, 主题建模, 情感, 大数据, 推特, 底特律

Introduction

The overall well-being of residents in cities that have experienced long-term and sustained population loss—so-called shrinking cities—is a critical matter of interest to urban planners. A good deal of scholarship indicates that urban shrinkage is often associated with reduced urban vitality—physically, economically, socially, and demographically (Gans 1975; Newman et al. 2018). For example, urban shrinkage can reduce tax revenue, which places a more significant tax

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¹Ewha Womans University, Seoul, South Korea

²Auburn University, Auburn, AL, USA

Corresponding Author:

Yunmi Park, Architectural and Urban Systems Engineering, Ewha Womans University, #311-2 Asan Engineering Bldg. 52 Ewhayeodae-gil, Sinchon-dong, Seodaemun-gu, Seoul 03760, South Korea.
Email: ymp@ewha.ac.kr

burden on the remaining residents for essential city services and infrastructure maintenance, property neglect and abandonment, and increased crime, poverty, and unemployment (Heim LaFrombois, Park, and Yurcaba 2019). While population loss does not necessarily negatively affect individual city residents' well-being (Hollander 2011), population loss and its related negative effects often lead to perceptions of the city as a whole being unhealthy and undesirable (Hollander et al. 2009, 230). Instead of the actual quality of places, the perception makes people avoid the area (Park and Garcia 2020), which could result in intentional disinvestment (King 2015).

To mitigate or outright avoid these potential negative consequences and perceptions, city leaders and planners often focus on urban regeneration. This process seeks to bring about a reversal of the current stagnation and decline. For instance, the Cleveland Foundation and the city administration of Cleveland, Ohio, invested more than \$44 million to revitalize two historical city neighborhoods (Wright, Hexter, and Downer 2016). Manchester, United Kingdom, also invested billions of pounds in rejuvenating two square miles of its city core (Reynolds and Leonard 2020). In addition, the federal government of South Korea developed a plan to invest almost \$46.5 billion in urban regeneration projects (Kim and Choi 2018). Despite the huge financial investments in urban regeneration projects worldwide, not all have produced "revitalized" cities or garnered the support of their residents. The successes and failures of these projects are complex and context-specific. However, the scholarship is increasingly seeing the value in the opinions of the people who live, work, visit, and play in the urban neighborhoods as a key factor in strategic and targeted approaches to urban regeneration (Maurrasse 2006). As such, and in line with advocates of participatory planning practices, the first step in any urban revitalization strategy is listening to the neighborhoods' residents.

While there is a great deal of scholarship on participatory approaches to urban planning, social media's role is emerging as critically important, especially during a global pandemic. We see limitations on in-person gatherings and residents simultaneously taking to the streets to protest and document via social media the systemic inequalities that are part of the fabric of our cities. Big data derived from social network services (SNS), such as Instagram, Twitter, or Pinterest, allow access to some of their data through an open API (application programming interface) and can be an effective means of pursuing social listening. As of 2019, there are an estimated 3.5 billion SNS users, and their numbers are growing by 9 percent each year (Basit, Nurlukman, and Kosasih 2020). As big data generated from SNS grow, they may become a valuable tool to assess public opinion and perceptions. While there are several limitations related to privacy, inconsistent data collection, and age-group bias (Hollander et al. 2016), these data are as valuable as those of conventional data collected by census or survey. SNS data

include long time horizons and large sample sizes and require less effort and time—all of which are significant concerns for leaders of shrinking cities with limited resources.

This research examined social media's role in understanding city residents' perceptions about the spaces within which they interact, what planners and policymakers can learn from social media data, and how to use the data to inform urban regeneration efforts through participatory means. Using Twitter data from 2010 to 2018 recorded in Detroit, Michigan, this paper investigates the topics people discussed regarding their city, their sentiments on the topics, and the associated neighborhood conditions. Detroit is among the fastest depopulating cities in the United States from 1980 to 2010 (Yin et al. 2020; Xie et al. 2018). While the city shows signs of recovery thanks to public and private investments, planners often consider Detroit a textbook example of a shrinking city, making it an ideal case for tracking changing trends in sentiments and the association of such trends with neighborhood conditions.

The aims of this research are to understand the changing nature of sentiments associated with the city and how these evolving sentiments are related to changing neighborhood characteristics and, using social media data, to identify a new approach, albeit an approach to complement and be used in conjunction with a vast array of other approaches, for improving neighborhoods through participatory, citizen-driven planning—for example, how neighborhoods can be identified for planning activities through people's sentiments as they change over time. This approach, which relies on SNS data from those directly interacting with a city's spaces, is intended to help planners and policymakers in shrinking cities in their urban regeneration efforts.

Urban Decline and Shrinkage Sentiments and Determinants

Urban Decline and Shrinkage

Urban decline is a general term used to describe an overall reduction in the quality of life within a city. While the causes and patterns of decline vary by city, a frequent sign of urban decline is population loss. Interrelated processes of population out-migration, deindustrialization, reduced local economic opportunities, and natural disasters characterize population loss (Fol 2012). Furthermore, demographic and household restructuring resulting from lower birth rates and an aging population escalates shrinkage in many developed countries (Heim LaFrombois, Park, and Yurcaba 2019). Regardless of the context-specific reasons for urban shrinkage in a given city, its negative impacts are well documented, such as the reduced provision of public services, increased crime, greater poverty, increased tax burden, and an abundance of vacant property (Bernt 2009; Rybczynski and Linneman 1999). These negative impacts can further reduce the city's overall quality of life and fiscal

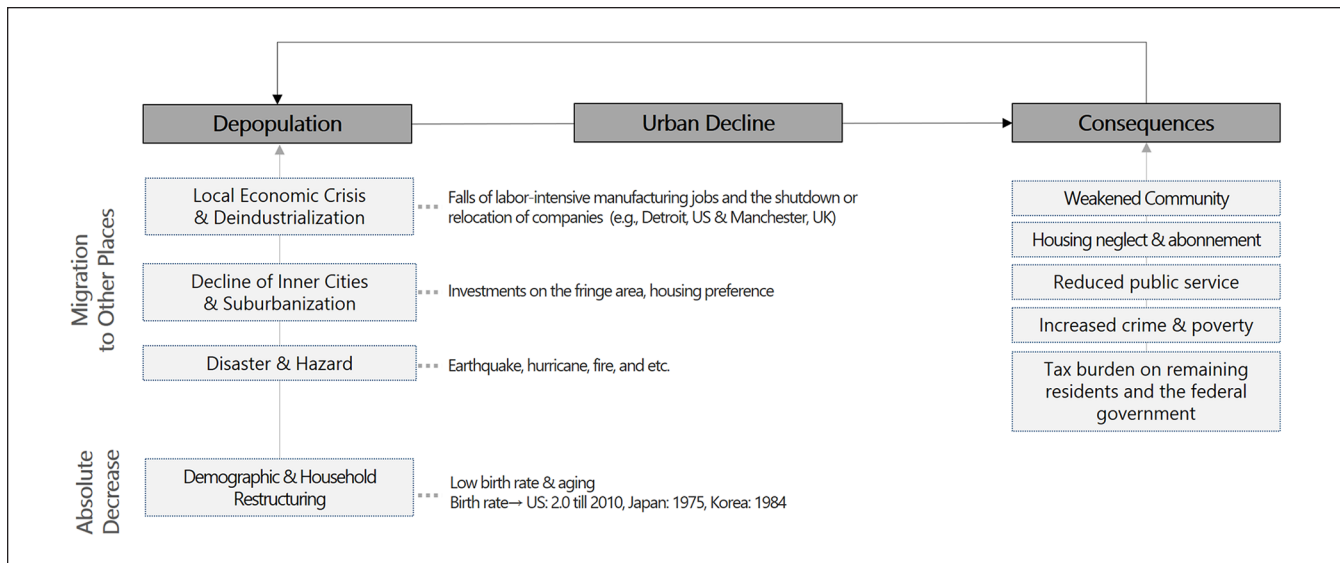


Figure 1. Consequences of urban decline.

health, contributing to ongoing population loss and intensified urban decline (Figure 1).

Given these negative consequences of urban decline and shrinkage, people perceive shrinking cities as unhappy, unhealthy, and undesirable (Y. Park and Heim LaFrombois 2019). However, empirical studies of the phenomenon have produced a mixed bag of results. For example, according to Hollander (2011) and Hollander and Renski (2015), a comparison of ratings for neighborhood satisfaction from the American Housing Survey and sentiment scores derived from Twitter in growing and depopulating cities does not present statistically significant differences between the sets. However, Mitchell et al. (2013) found that “happy” words are more likely to appear in growing cities in the United States than in other cities. There are, of course, exceptions—researchers found several cities in Texas to be “unhappy” but growing in population. Meanwhile, Delken (2008) found that average life satisfaction is not statistically significantly different between shrinking and growing cities in Germany; instead, the important factor is living in West or East Germany. While research continues to emerge that compares sentiments in shrinking cities to non-shrinking cities to understand whether urban decline and shrinkage affect happiness levels, what is clear from this ongoing scholarship is that perceptions of a place are important.

Sentiment Analysis and Its Determinants

Sentiment analysis is a widely used method for extracting emotions or feelings from text data (Calvo and D’Mello 2010; Cambria et al. 2017). In its simplest form, it codes language and text effects as either positive or negative. Researchers can classify negative sentiments into several categories: anger, anxiety, sadness, envy, fear, and disappointment, and each

can be studied separately (HUMAINE 2006). Positive sentiments have similar subtypes, such as satisfaction, happiness, and subjective well-being.

Previous studies also examined demographic characteristics (e.g., gender, age, ethnicity, marital status); socioeconomic status (e.g., income, education, poverty, unemployment); physical and psychological health; social relationships with friends, neighbors, or partners; and neighborhood conditions as possible determinants of sentiments. Among this scholarship, demographic characteristics show a range of different impacts on sentiments. For example, some found that women express more negative sentiments than men (Green and Sabini 2006), whereas others reported the opposite (Santiago, Wadsworth, and Stump 2011). Meanwhile, Deeming (2013), using U.K. data, found no significant effect on emotional expression between men and women. Researchers also found that age had both positive and negative effects on sentiments, and several studies found that older people express less negative sentiments than younger people (Jakoby 2016; Mankus, Boden, and Thompson 2016). In addition, being a member of a minority group and being unmarried tend to increase negative sentiments (Clark et al. 2017; Deeming 2013; Gallo et al. 2005).

Higher unemployment, greater poverty, lower educational attainment, and lower incomes are all likely to increase negative sentiments (Diener et al. 2010; HannsScott 2016; Kushlev, Dunn, and Lucas 2015; J. Park et al. 2013; Petersen 2010). However, these findings do not always hold when dividing positive and negative sentiments into more specific categories. For example, Kushlev, Dunn, and Lucas (2015) established a relationship between income and the negative emotion of sadness but found no definite relationship between income and the positive emotion of happiness. Poor physical health, psychological health, and

social relationships increase negative sentiments, and people with physical or mental illness who have unsatisfied psychological needs express more negative sentiments (Clark et al. 2017; Diener et al. 2010). Lower life satisfaction and worse social relationships increase negative sentiments (Bastian et al. 2014; Jakoby 2016).

Researchers also found that neighborhood conditions influence sentiments. For example, housing value, which possibly reflects the quality of nearby amenities of neighborhoods; openness or proximity to nature, green space, and open space; accessibility to private and public transportation; and neighborhood design and planning features, including population density, land use, housing types, and street patterns, all impact sentiments (Cloutier, Jambeck, and Scott 2014; Farrell, Aubry, and Coulombe 2004; Guest and Lee 1983; Larson, Jennings, and Cloutier 2016; Soukiazis and Ramos 2016). In most cases, better neighborhood conditions (e.g., higher housing values, greater openness or greening, and better accessibility) tend to increase the chances of positive emotional expression.

Research Gap and Questions

Demographic and socioeconomic characteristics, health, social relationships, and neighborhood conditions are major determinants of sentiments, but the specific findings are mixed and qualified over the literature as a whole. Potential reasons for these mixed results could include different study settings, sample sizes, characteristics of the survey respondents, data types, units of analysis, and study areas (Y. Park 2017). Pursuing additional studies and obtaining a more diverse data set can benefit the understanding of sentiments and their determinants. In this regard, the urban planning field is increasingly using sentiment analysis with information derived from SNS data as vast amounts of big data from SNS are publicly available and are often free. In addition, automated sentiment analysis techniques have made it possible to mine large volumes of sentiments in short amounts of time (Feng et al. 2018). Thus, sentiment analysis with SNS is gaining attention as an essential method for understanding people's attitudes, opinions, and sentiments, allowing planners and policymakers to respond quickly and appropriately. For example, researchers found hot spots of happiness around green space (Lim et al. 2019) and commercial and public areas (Cao et al. 2018). However, researchers have not satisfactorily explored the association between the condition of neighborhoods and sentiments and how these associations have been changed in shrinking cities. It is at this juncture that this research intervenes.

While several studies compared sentiments from cities' residents and visitors using data derived from SNS, these observations used data within a limited period. For instance, Mitchell et al. (2013) took observations from one year, Nguyen et al. (2016) reviewed data for six months, Plunz et al. (2019) for eighteen months, and Quercia et al. (2012) for fifteen months. While cross-sectional studies are essential for

offering a snapshot of a given point in time, long-term exploration is better suited for identifying those factors that have constant or changing impacts on sentiments over time, such as in cities where change is inevitable. A detailed, longitudinal case study of neighborhoods in shrinking cities, in particular, is necessary to understand changes in and determinants of sentiments to a sufficient degree to develop necessary planning actions.

As such, this research investigates one shrinking city—Detroit, Michigan—longitudinally to understand the changing nature of sentiments associated with the city and how these evolving sentiments relate to changing neighborhood characteristics. Specifically, we examine three research questions relating to changes over time:

Research Question 1: What topics are people discussing regarding their city?

Research Question 2: What are their sentiments and perceptions on these topics?

Research Question 3: What neighborhood conditions are associated with these topics and sentiments?

In other words, what kinds of topics and associated sentiments emerge and disappear over time in Detroit's neighborhoods, and what neighborhood characteristics affect these expressions of sentiments over time? As we examined these questions, we also explored social media's role in understanding people's perceptions about the spaces within which they interact. Our study informs planners and policymakers about social media data, including urban regeneration strategies through participatory means.

Research Design

Study Area and Timeframe

Detroit, Michigan, is a textbook example of a shrinking city. Its characteristics include significant population loss, abundance of abandoned and dilapidated properties, underfunded and underused public infrastructure, persistent unemployment, and high crime rates (McDonald 2014). The causes of Detroit's decline are complex and far-reaching: an economy dominated by a single industry, the automotive industry, was severely affected by the downsizing and relocation of the industry, tense race and labor relations, suburbanization and white flight, and problematic political leadership (McDonald 2014). The city's population has been declining since its peak in 1950; however, a sharp fall was experienced more recently between 2009 and 2010 when the city lost 1.1 million residents (Manson et al. 2019). In 2013, the city filed for bankruptcy, making it the largest city in the United States ever to do so (Koven and Koven 2018).

Yet, despite a long trajectory of decline, the city has exhibited signs of recovery since 2010: improving employment rates, gains in gross domestic product per capita, and increasing median property values (McDonald 2014). In 2010, Quicken

Loans moved its headquarters to Detroit. In addition, JP Morgan's Corporate Social Responsibility (CSR) program, inaugurated in 2014, provided support for urban regeneration efforts, including job training programs, construction of affordable housing, conversion of vacant properties into shared workspaces, and downtown revitalization projects (Koven and Koven 2018). Michigan Central Station is slated for renovation by the Ford Motor Company to house some of its workers (Austin 2018), and several new downtown developments are ongoing (City of Detroit (MI) Planning and Development 2020). Our study examined the timeframe between 2010 and 2018, including positive and negative occurrences within the city that could be associated with changing neighborhood conditions and residents' and visitors' sentiments.

Data Collection and Analysis

This research examined Twitter postings reported in Detroit from 2010 through 2018 to identify the sentiments relating to specific areas of the city or neighborhoods. We also looked at how these sentiments change over time and concerning neighborhood conditions. We conducted three research processes: sentiment scoring, topic modeling with Word2vec, and associating sentiments with neighborhood conditions using panel modeling to investigate the topics that appear and disappear over time and the sentiments and neighborhood characteristics of these topics. We used census tracts as operational surrogates of neighborhoods since census tracts well represent institutional neighborhoods, which provide a wider variety of land use leading to different activities and where planning interventions could occur (Y. Park and Rogers 2015).

We derived the sentiment score from Twitter postings due to advantages over other SNS platforms: it is large and mainstream (Hollander et al. 2016), particularly in the United States, with over 68 million users as of 2019 (Pew Research Center 2019). Earlier studies also used Twitter, which ensures validity and cross-comparability of the data. Also, Twitter allows users access to the data through API, unlike Facebook and Instagram. We only collected geocoded data. We removed transmissions from bot accounts, which can generate a vast amount of Tweets, by calculating the complete automation probability (CAP) score (i.e., $CAP > 0.5$) as guided by Juhász and Hochmair (2019). This process reduced the total number of tweets in the data set from 958,763 to 908,992. Although previous literature like Lansley and Longley (2016) found geo-tagged data mostly illustrate geo-tagged places—tweets related to museums and galleries presented high densities at museums—tweets including “ford field” and “traffic” were filtered and mapped as to make sure those data described the neighborhoods where they were nested (see Supplemental Appendix 1 for the maps). Tweets including “ford field” mostly were posted around the “ford field” and “traffic” along the roads. Then, each tweet's sentiment as positive, neutral, or negative was identified using the AFINN lexicon dictionary developed by Hansen et al. (2011). Several lexicon

dictionaries are in use, but relevant studies used the AFINN for sentiment analysis (Lowe et al. 2011; Markovikj et al. 2013). In particular, Koto and Adriani (2015) argued that this lexicon dictionary is best for analyzing sentiments in Twitter data. AFINN provides a list of English terms, and scores coded words with an integer between -5 and $+5$. Here is an example of positive, neutral, and negative tweets. “My mama fucking annoying” got -6 points: -4 from “fucking” and -2 from “annoying.” Because we also aggregated the sentiment scores at the census tract level, we used three different sentiment scores: the ratio of positive tweets over total tweets (PR), the ratio of negative tweets over total tweets (NR), and relative sentiment score (RSS). We calculated the RSS using the relative proportion of total numbers of positive (p) over negative tweets (n), following the guide of Lowe et al. (2011): $RSS = (p - n) / (p + n)$.

We conducted topic modeling using the Word2vec method developed by Mikolov et al. (2013). It is a popular text analysis method that understands word associations and detects a cluster of synonymous words using a trained neural network. Word2vec has two model types: continuous bag-of-words (CBOW) and continuous skip-gram model (skip-gram). In this study, we adopted skip-gram because of its superior performance of semantic analysis over CBOW, particularly when there were many infrequent words. We then grouped and analyzed words using k-means clustering and mapped through “Scikit-learn” and “Seaborn” libraries in python (Waskom 2013). We determined the number of clusters, $k = 5$, at the point where the inertia value rapidly decreases—inflection point (see Supplemental Appendix 2 to find the inflection points and graphs)—as guided by Pedregosa et al. (2011). We explored the topic variations every year and named each group's title based on the subsets of words that included each cluster's title. Although the titles could not catch all subsets of words, they represented an overall picture of the word clusters.

Next, we observed the association between sentiments and neighborhood conditions, each collected and analyzed at the census tract level (we initially identified 311 tracts, but we excluded eight due to missing values). We grouped the variables under the constructs of demography, socioeconomic status, and the built environment. We retrieved data on neighborhood conditions from the American Community Survey's (ACS) five-year estimates. Poverty, income, and several other socioeconomic variables were not included in the final model due to multicollinearity issues with other socioeconomic variables.

We conducted a factor analysis to create a scale that embraced different socioeconomic status variables into a few, but we could not constantly reduce those variables into one or two variables across the years; factor loading changed from year to year. Crime data, collected from the Detroit Data Portal, were excluded because it did not cover the entire period. We collected population density, years built, and vacant home variables to identify the built environmental traits. Although the North American Industry Classification System informed the neighborhood's business diversity

Table 1. Variables of Interest and Descriptive Statistics (2010–2018).

Construct	Variables	Measures	N	Mean	SD	Range
Demography	Youth ratio	Ratio of people under 18 years old	2,753	0.25	0.08	0–0.51
	Female ratio	Female ratio	2,753	0.52	0.06	0.21–1
	African American ratio	Ratio of African American	2,753	0.81	0.26	0.00–1
	Single ratio	The ratio of single over married	2,739	3.13	3.01	0.20–89.33
Socioeconomic status	Unemployment rate	Unemployed people	2,753	0.11	0.06	0–0.75
	Less education	Ratio of people with less than high school diploma	2,753	0.14	0.07	0, 0.41
	Housing value	Median housing value (\$)	2,653	57,598.53	41,362.44	9,999–467,900
	Income	Median income (\$)	2,746	27,521.37	11,728.54	8,733–131,518
Built environment	Population density	Number of people per square mile	2,790	5,741.14	2,924.16	0–20,243.6
	Years built	Average building age (years)	2,655	66.77	9.58	13–78
	Vacant housing ratio	The ratio of vacant over occupied housing	2,753	0.48	0.31	0–2.22

changes, we provided the data by ZIP code. We did not find a nice match between the census tract and ZIP code boundaries in Detroit. Therefore, the interpolation induced another concern, especially non-count variables such as median income, median years built, and average housing value (Liu and Martinez 2019), and reduced sample size. Thus, we excluded the land-use change variable from the model. We included a year dummy to control the influence of unidentified events each year. This inclusion accounted for the possibility of certain events evoking streams of communication through SNS. Table 1 lists the final variables used in the analyses.

Finally, a standard fixed-effects (FE) model was run with strongly balanced panel data with 303 observations. The Mundlak test was conducted instead of the Hausman test to determine whether a fixed model was to be preferred over a random model ($\chi^2 = 82.72, p = .00$).

Results

Descriptive Statistics

The initial examination of long-term sentiment data showed several interesting facts (Table 1; Figure 2). PR was fairly stable relative to NR, which continually declined until 2014 and did not rebound to the level of 2010 until 2017. PR maintained the upper hand and was relatively stable, except in 2014 and 2015. RSS, like NR, fell—RSS was negative in 2013—which indicated that negative expressions were more common than positive ones. The range of PR and NR by tracts also dramatically widened after 2016, implying greater variance in neighborhoods' sentiments.

As shown in Figure 2, over the nine-year study period, Detroit also experienced changes in socioeconomic status, built environment, and some demographic features. The mean values for age, gender, education level, and proportion of African American residents remained relatively constant over time in all tracts. The ratios of employed to unemployed and single to married, and real estate values exhibited visible changes. Unemployment reached a peak (20% in an average

tract) in 2010 and improved steadily until 2018. More than three times as many singles as married couples are present by 2018, and the single population showed constant growth. Mean and median housing values fell until 2015 but have been increasing ever since. The average age of buildings fell over time, which may indicate new developments or the demolition of vacant housing; the rate of vacant housing increased through 2010 and stabilized after 2017. Housing values have increased since 2016, and the age of buildings fell after 2016, again possibly due to the removal of old vacant properties or new construction.

Topic Clusters: The Results of Word2Vec

We identified five different clusters of topics to detect the few clusters of vital posts in Detroit while not breaking down all tweets with great details. Table 2 presents the topics we found from tweets each year, which have no ranking order. The following clusters appeared across all observation years: “Weather (e.g., thunderstorm, snow, or sunny),” “City & Places (e.g., parks, bar, or market),” and “Event & Entertainment (e.g., music, party, concert, or drink).” However, only in 2015, “Weather”-and “Drug”-related words (e.g., smoke, weed, marijuana, or cannabis) appeared. Similarly, a cluster of “Slang” disappeared after 2015, while optimism appeared in 2016. Interestingly, “Job Search” appeared in 2011.

Determinants of Sentiment

Table 3 presents the result of the FE models. The NR model explains more variance than the PR model (adjusted R^2 values were 18.24% and 6.69%, respectively), while the RRS model had the highest explanatory power of 22.44 percent. This finding implies that the variables of interest could explain negative sentiments more than positive sentiments.

The PR model suggests that a census tract with a greater single ratio over more expensive houses is likely to show more positive sentiments, statistically significant at the .1 level. The

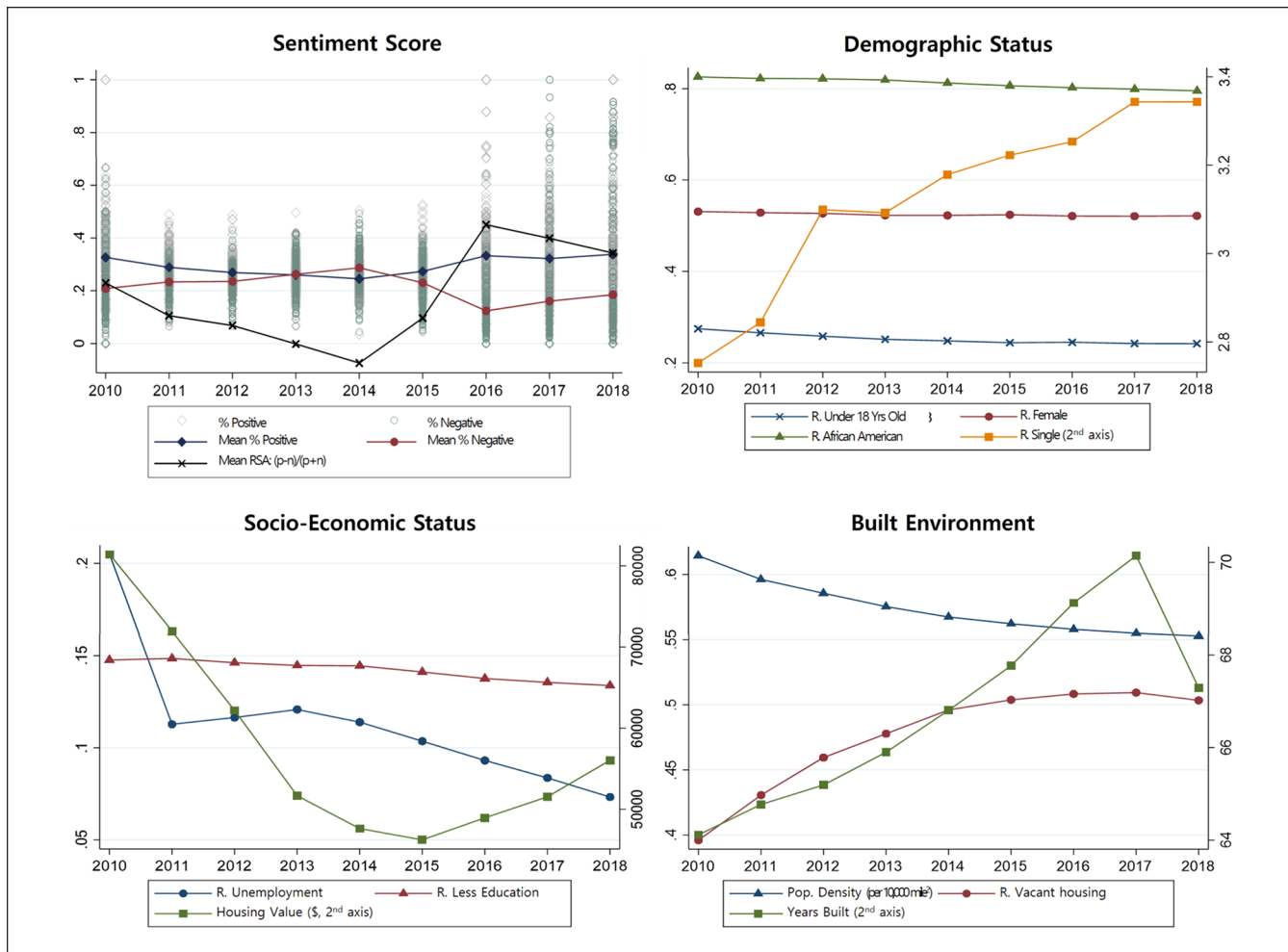


Figure 2. Mean changes in variables by year.

Table 2. Topics from Years 2010–2018.

2010	Sports & Event 2,847	Slang 858	City & Place 390	Shopping & Entertainment 279	Weather 162
2011	Sports & Event 7,539	City & Place 1,070	Slang 2,504	Job Search 320	Weather 195
2012	Event & Entertainment 5,998	Daily Life 2,601	Slang 1,204	City & Place 1,196	Weather 201
2013	Event 6,464	City & Place 1,985	Slang 1,629	Sports & Event 610	Weather 209
2014	Event & Entertainment 5,579	Daily Life 2,366	City & Place 2,205	Slang 927	Weather 200
2015	Mixed 9,440	Slang 2,228	City & Place 2,147	Public Facilities 1,529	Weather & Drug 335
2016	Cultural Figures 10,112	Optimism 3,826	City & Place 1,845	Event & Entertainment 1,225	Weather 358
2017	Creative/Arts 9,094	Optimism 2,966	City & Place 1,842	Event & Entertainment 1,052	Weather 339
2018	Event & Entertainment 6,223	Optimism 2,100	City Place 941	Weather 334	Traffic & Accident 170

Numbers under each topic refer to the total word counts of nested words under each topic.

Table 3. Outcomes of Fixed-Effects Models.

Variables	Positive ratio		Negative ratio		RSS	
	Coef.	p value	Coef.	p value	Coef.	p value
Youth ratio	-.092	.18	.136	.04	-.423	.03
Female ratio	.076	.26	.069	.29	-.024	.90
African American ratio	-.059	.32	.165	.00	-.281	.10
Single ratio _(logged)	.014	.08	-.014	.07	.031	.17
Unemployment rate	.071	.27	-.138	.02	.398	.03
Less Education	-.041	.62	.158	.04	-.376	.11
Housing value _(logged)	.017	.10	-.019	.05	.078	.01
Income _(logged)	.024	.14	.003	.84	.003	.94
Population density ^a	.006	.19	-.004	.39	.026	.04
Years built	.000	.87	-.001	.34	.002	.39
Vacant housing ratio	0.021	0.33	-0.009	0.68	0.069	0.27
Year-2011	-.028	.01	.012	.24	-.080	.01
Year-2012	-.046	.00	.014	.19	-.112	.00
Year-2013	-.053	.00	.043	.00	-.179	.00
Year-2014	-.067	.00	.067	.00	-.247	.00
Year-2015	-.039	.01	.009	.48	-.075	.06
Year-2016	.023	.11	-.095	.00	.285	.00
Year-2017	.012	.42	-.061	.00	.229	.00
Year-2018	.035	.03	-.047	.00	.197	.00
Constant	-0.126	0.58			-0.696	0.28
Adjusted R ²	6.69		18.24		22.44	
No. of census tracts	303		303		303	
No. of tracts, all years	2,542		2,542		2,528	

Note: RSS = relative sentiment score.

^aMultiplied by 1,000.

NR model finds a similar effect for singles and housing values as the PR model. In addition, the NR model indicates that a tract with more youth, more African Americans, lower unemployment, and lower education levels tends to have more negative sentiments. The RSS model incorporates the combined results of the PR and NR models, adding that tracts with higher population densities have relatively positive sentiments. In all models, age and housing value had an impact on sentiments—meaning, the older the age of the individual and the more expensive the housing, the happier the residents. The year variable in the RSS model and some years in the PR and NR models have a statistically significant impact. This result implies that an annual fluctuation or certain events impacted the sentiment scores more than the other included variables.

Discussion

Several findings from this research may help inform and guide the work of planning academics and practitioners alike, even if more studies and work are required.

Our research aimed to understand the changing nature of sentiments associated with the city and how these evolving

sentiments relate to changing neighborhood characteristics. Namely, what kinds of topics and associated sentiments emerge and disappear over time in Detroit's neighborhoods, and what neighborhood characteristics affect these expressions of sentiments over time? Using topic clustering, planners may easily identify major places and activities that interest people, such as sports, events, shopping, entertainment, and public facilities. Topic clusters could aid in tracing the trends of people's interests, so planners could carefully look into these places and community events. Moreover, it is found that "bottom-up" factors, such as sociodemographic traits, environment, and external events, influence hedonic happiness; so-called "top-down" factors, regarding personality, individual activities, and neighboring, determine sentiments heavily though (Y. Park, Kim, and Seong 2021). The lower explanatory power of the three panel models supported that other unidentified factors could explain the level of sentiments.

The results from this research suggest that neighborhoods with a younger population tended to report relatively more negative feelings, a finding supported by the existing scholarship (Jakoby 2016; Mankus, Boden, and Thompson 2016). Age differences in response to problems and negative

sentiments are well studied. Charles and Carstensen (2008) concluded that older adults could regulate their emotions better and give less expression to their anger. Reis and Rusbult (2004) found that the elderly tend to avoid negative stimuli. Or this only could be true since younger and more active populations post high volumes, which would dominate the tweets (D'Avanzo, Pilato, and Lytras 2017).

In addition, this research found that neighborhoods with higher proportions of singles versus married individuals were more likely to report positive sentiments, which somewhat counters the existing scholarship, like DePaulo and Morris (2006) who reported that singles were less happy than married couples. Reis and Rusbult (2004), however, argued that married couples are not always happy over the long term and that single individuals are slightly less happy with an only 0.2-point gap on an 11-point scale than people who get married and stay married. Connecting this finding to the field of planning, many recent urban regeneration projects are concentrated in downtown areas, which are environments that are better suited to singles and their consumer preferences. These projects, in turn, may have resulted in more positive sentiments from singles because the built environment was developed with their demographic in mind. In other words, public transportation, new mixed-use developments, bars, concert halls, and sports arenas developed in these areas could stimulate a sense of positivity and social media posting from this specific demographic.

Neighborhoods with fewer African Americans, more educated individuals, and more expensive housing, which when taken together could imply a "better" neighborhood, tended to report more positive sentiments, which echoes previous studies. Cutrona, Wallace, and Wesner (2006) reported that disadvantaged neighborhoods are more vulnerable to negative events and host disrupted social ties. Unpleasant physical surroundings may increase daily stress levels, resulting in negative emotions, whether it is recognized or not (Heckert and Mennis 2012). Counterintuitively, higher unemployment rates tended to result in higher sentiment scores. A few previous studies produced similar results, arguing that many unemployed people find happiness through their friends and families, religious communities, or events occurring in their everyday lives (Myers 2000). Santiago, Wadsworth, and Stump (2011) found a positive impact of unemployment on sentiment due to decreased work stress. As noted above, Kushlev et al. (2015) found no definite relationship between income and happiness. Knabe et al. (2010) found that the unemployed adapt to changing life circumstances and seek ways to use their free time to yield higher satisfaction levels. Since Twitter does not provide information on other correlates, such as religion, friendship, or life events closely tied to sentiments, future studies should examine these factors to improve generalizability.

SNS posting is often related to presenting perspectives that reflect self-worth, self-absorption, or narcissism (Weiser 2015). As such, positive posts are typically more common

than negative ones (Utz 2015) if postings are related to their daily lives or personal things. However, negative emotions dominate political behavior and public opinion-related postings (Barberá 2016). And negative sentiments from negative events elicit stronger and quicker emotional responses than positive ones (Stieglitz and Dang-Xuan 2013). It is also found that specific local phenomena and topics set the sentiment orientations and emotions in social media. For example, the NR exceeds PR in 2014 in Detroit. Moreover, the coefficient of year dummy in the fixed models showed a maturation effect (i.e., negative posting increased, while the positive posting decreased relative to 2010) and a history effect (i.e., a reversed slope for 2014–2016). While the exact causes of these effects are unknown, several noteworthy things took place during this time that could be a factor in the reversal of trends. Combining results from topic modeling, a cluster of optimistic words emerged, while the slang cluster disappeared in 2016. We are not sure what caused the sentiment changes, but there were some incidences around that time in Detroit. For instance, Detroit filed for Chapter 9 bankruptcy on July 18, 2013, and on November 7, 2014, received case approval (Koven and Koven 2018). Bankruptcy is generally perceived as a negative consequence and reflects poorly on the city's reputation and its ability to manage its fiscal affairs. However, such action also helped the city ease its debt burden and develop a reorganization plan that called for \$1.7 billion in revitalization projects to address blight, public safety, public transportation, and information technology system enhancements (Detroit Planning and Development 2014). This plan has led to developing a "blight database" (i.e., Motor City Mapping), creating a land bank, and removing blighted properties (Gallagher 2014).

Along with municipal efforts to shift the focus of planning from a growth machine to improving quality of life (Schindler 2016), large urban regeneration investments and Detroit Future City (2019) optimistic "Detroit Reinvestment Index" signal hope for the city's future. According to David, Ong, and Legara (2016), positive sentiments often appear on SNS after a disaster hits, and those postings include feelings of inspiration and expressions of support and hope. Therefore, possibly the city's bankruptcy and its resulting actions overall resulted in decreased negative sentiments. However, at this same time, residents expressed concerns about increasing living costs, new neighbors, displacement resulting from revitalization, and other issues related to gentrification (MacDonald and Chambers 2018). These new concerns appeared beginning in 2016, when the outcomes of the investments and plans for improvements may not have met expectations. This finding addresses the second aim of this research: social media's role in understanding people's perceptions about the spaces where they interact and what planners and policymakers can learn from the perceptions, explicitly relating to urban regeneration strategies. In this case, it was about displacement through gentrification, providing an opportunity for planners to intervene and engage

the community before confirming residents' worst fears. In addition, the silent majority may be likely to vent their emotions through social media before they express negative sentiments in action. As previously noted, high profile movements (e.g., #blacklivesmatter, #stopasianhate, or #stop hate) use social media platforms to organize collective actions and elicit stronger engagements (Vromen, Xenos, and Loader 2015). Thus, listening to social media could prevent disastrous events by taking actions beforehand or serving as an effective means of participatory actions.

While some critical findings have resulted from this research, there are several points to note. Perhaps most important is the use of Twitter data. As well documented, not all individuals or groups of individuals have equal access to technology. Known as the digital divide, this phenomenon "refers to the difference between those who can and have the opportunity to participate, compete, and prosper in knowledge-based economies and a society organized around social networks and those who do not" and in which age, race, and socioeconomic status determine access (Gordo 2002, 4). Therefore, it is quite possible that those who are younger, are not members of a racial minority group, and/or have a higher socioeconomic status are more frequent and prolific users of social media platforms, such as Twitter, although it is a free service.

A survey by Duggan and Brenner (2013) found differences regarding age, gender, and race by the type of SNS. The report states that 67 percent of Internet users are signed up to any SNS, but people in the 18–49 age cohort, women, and urban residents were likely to use SNS actively. The report found that Facebook users were more likely to be women aged between eighteen and twenty-nine years, while Twitter users were African Americans and urban residents aged eighteen to twenty-nine years. Following another report by Greenwood, Perrin, and Duggan (2016), they found that older adults joined in increasing numbers, but 21 percent of all U.S. adults are Twitter users. These somewhat educated and young adults (18–29: 59%, 30–49: 33%, 50–64: 18%, 65+: 8%) lived in urban settings. Singh, Halgamuge, and Moses (2019) stated that the variation in age and gender differences in Twitter users was regional, while Facebook users were more balanced in gender, education, income, and urban-rural areas.

Although some reports investigated who uses social media and the types of social media used—through survey or estimation using different computation approaches—few studies observed the types of people reacting on SNS and during what sorts of events specifically. This lack of research is mainly because identifying user information from social media is a barrier since the profile is neither public nor perfectly correct. For example, pioneering studies attempted to distinguish local and non-local residents (e.g., Kovacs-Gyori et al. 2018; Longley, Adnan, and Lansley 2015; Mislove et al. 2011) to help planners understand the needs of

residents and visitors. However, the methods of identifying local Twitter users are not matured yet (see Supplemental Appendix 3 for the difference between residents and non-residents). Future studies may develop more precise methods to discern residents from non-residents or identify socioeconomic and demographic characteristics under the interdisciplinary cooperation between planners and computer scientists. This information would help understand the differences in residents' or visitors' reactions to neighborhood characteristics or socioeconomic and demographic features.

Finally, the models were limited in the built environment-related variables due to data availability and consistency over long-term observation. Identifying land use diversities could inform the characteristics and change of neighborhoods that may be highly related to social media postings and emotions. However, the researchers retrieved most of the data from the census, so it was not possible to include land use changes that would shape activities and behaviors. The case studies in other cities that updated neighborhood condition data consistently, especially neighborhood's built environments relating to crime, land-use change, or major redevelopment projects, would increase the power of proposed models. Future studies could investigate other cities and compare the common factors that determine sentiment and the unique features of certain cities for better generalizability. Moreover, future research could diversify using data from different social media platforms other than Twitter, illustrating different perspectives due to different user groups. Moreover, future studies could extract tweets relating to specific issues (e.g., gentrification, living condition, or crime) and explore the sentiment to build a more refined picture based on our study's preliminary steps.

Conclusion

While the public health and psychology disciplines have well-documented effects of neighborhood conditions on physical and mental health, our research examined these issues from a planning perspective and longitudinal manner that considers how sentiments evolve in changing neighborhood conditions. It also highlights the role of social listening, in this case through social media, in the planning process, specifically related to urban regeneration efforts. Planners could cluster topics that people frequently tweet to understand people's concerns and find the association between neighborhood conditions and sentiments generated from these neighborhoods using Twitter updates. Both could help determine neighborhoods' occupant feelings and desires. This research supports the findings from previous non-planning-based research in that neighborhood demographic, socioeconomic, and built environment conditions impact people's emotions and sentiments. Several noteworthy events in Detroit during the timeframe of study roughly map onto these trends. The effects of large-scale planning and

policy interventions on neighborhoods may influence overall perceptions and sentiments of a city's populous. These results are particularly interesting to planners and public officials in shrinking cities, such as Detroit, for several reasons. First, there is concern about the perception of a city as an unhealthy and undesirable place to live and do business. If city's residents perceive their city negatively, it could lead to further depopulation. Planners could use the city residents' perceptions to gauge larger sentiments shared across the region or country. Second, as leaders of shrinking cities invest in plans and initiatives to reverse depopulation and disinvestment trends, they need their residents' support.

The policy implications of this research are twofold, relating to the planning process and planning practice. Using social media to gather input from a community's residents is a widely used practice. However, employing social media feedback as a process of "social listening" can influence the shape of the engagement process. Social media provides a more significant opportunity to learn about residents' concerns while reducing the participation barriers of more structured, inflexible, and formal forms of public engagement: closed or semi-closed response surveys, town hall meetings with limited comment opportunities, and extensive in-person workshops. For example, planners would not need public meetings and hearings at inconvenient times in unchecked places. No prompting questions would direct responses toward a particular issue, no "educating" individuals on the issues and possible solutions, and no reacting to comments; all they need is listening. Through listening, planners and city leaders can begin to identify areas of focus based on residents' priorities, determine levels of support for various planning interventions/investments, or learn what in-depth conversations and engagements need to take place. In light of global protests occurring at the time of this writing, it is clear that listening to people is at the forefront of thoughtful planning.

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ORCID iDs

Yunmi Park  <https://orcid.org/0000-0002-7924-6518>

Jiyeon Shin  <https://orcid.org/0000-0002-3331-399X>

Megan E. Heim LaFrombois  <https://orcid.org/0000-0002-2196-0655>

Supplemental Material

Supplemental material for this article is available online.

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Author Biographies

Yunmi Park, PhD, AICP, is an assistant professor at Ewha Womans University, and her specializations and research interests are in urban planning and policy, urban shrinkage and revitalization, land-use planning, and spatial analysis.

Minju Kim is a graduate student in the Architecture and Urban Systems Engineering of Ewha University in South Korea. Her interests are in spatial analysis, data mining, and machine learning.

Jiyeon Shin is a graduate student in the Architectural and Urban Systems Engineering of Ewha Womans University, and her research interests are in environmental planning, environmental justice, and community development.

Megan E. Heim LaFrombois, PhD, AICP, is an associate professor in the Department of Political Science, Community Planning Program, at Auburn University in Auburn, AL. Her research focuses on community development, participatory urban planning, feminist approaches to urban studies and research, public policy, and urban inequalities.